OS LAB-2
DATE: 2/3/26

SUMANT SHRIDHAR
USN: 1BM24CS299
BATCH: 4K2

1) FCFS:

```c
#include <stdio.h>

int main() {
    int n;
    float avgWT = 0, avgTAT = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int AT[n], BT[n], CT[n], TAT[n], WT[n];

    for(int i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time of P%d: ", i+1);
        scanf("%d %d", &AT[i], &BT[i]);
    }

    CT[0] = AT[0] + BT[0];

    for(int i = 1; i < n; i++) {
        if(CT[i-1] < AT[i])
            CT[i] = AT[i] + BT[i];
        else
            CT[i] = CT[i-1] + BT[i];
    }

    for(int i = 0; i < n; i++) {
        TAT[i] = CT[i] - AT[i];
        WT[i] = TAT[i] - BT[i];
        avgWT += WT[i];
        avgTAT += TAT[i];
    }

    printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");
    for(int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               i+1, AT[i], BT[i], CT[i], TAT[i], WT[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avgWT / n);
    printf("\nAverage Turnaround Time = %.2f\n", avgTAT / n);

    return 0;
}
```

OUTPUT:

```
Enter number of processes: 4
Enter Arrival Time and Burst Time of P1: 0
5
Enter Arrival Time and Burst Time of P2: 1
3
Enter Arrival Time and Burst Time of P3: 2
8
Enter Arrival Time and Burst Time of P4: 3
6

Process AT      BT      CT      TAT     WT
P1      0       5       5       5       0
P2      1       3       8       7       4
P3      2       8       16      14      6
P4      3       6       22      19      13

Average Waiting Time = 5.75
Average Turnaround Time = 11.25
```

Lab 2

2/3/26

1) Write a C program to simulate the following CPU scheduling algorithms to find the average TAT and average WT.

a) <u>FCFS</u>

```
#include <stdio.h>

int main(){
    int n;
    float avgWT = 0, avgTAT = 0;

    printf("Enter number of processes : ");
    scanf(" %d", &n);

    int AT[n], BT[n], CT[n], TAT[n], WT[n];

    for (int i=0; i<n; i++){
        printf("Enter AT, BT for process %d : \n",
                                        i+1 );
        scanf("%d %d", &AT[i], &BT[i]);
    }

    CT[0] = AT[0] + BT [0];

    for (int i = 1 ; i<n ; i++){
        if (CT[i-1] < AT [i] ){
            CT[i] = AT[i] + BT[i].
        }

        else CT[i] = CT [i-1] + BT[i]
    }
```

```c
for (int i = 0; i < n; i++) {
    TAT[i] = CT[i] - AT[i];
    WT[i] = TAT[i] - BT[i];

    avgWT += WT[i];
    avgTAT += TAT[i];
}

printf("\nProcess\tAT \tBT \tCT\tTAT\tWT\n");

for(int i = 0; i<n; i++){
    printf("P%d \t%d \t%d \t%d\t%d\t%d
        \n", i+1, AT[i], BT[i], CT[i],
        TAT[i], WT[i]);
}

printf("Average TurnAround Time : %.2f \n",
        avgTAT/n);

printf(" Average Waiting Time : %.2f \n",
        avgWT/n);

return 0;

}
```

03/08/25

O/P :

```
Enter number of processes: 4
Enter AT, BT for process 1:
0
5

Enter AT, BT for process 2:
1
3
```

Enter AT, BT for process 3 :
2
8
Enter AT, BT for process 4 :
3
6

Process	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	8	16	14	6
P4	3	6	22	19	13

Average Turn Around Time : 11.25
Average Waiting Time : 5.75